PAPER_689: AGN Relativistic Jets: Blandford-Znajek Mechanism and UQFF Modulation

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Abstract

Active galactic nuclei (AGN) launch relativistic jets powered by the Blandford-Znajek mechanism — the extraction of rotational energy from a spinning black hole via magnetic field threading. This paper derives the jet power formula and UQFF-modulated acceleration.

1. Blandford-Znajek Jet Power

$$P_{BZ} = \kappa_{BZ} \frac{a^2 B^2 r_g^2 c}{4\pi}, \quad \kappa_{BZ} \approx 0.044$$

where $r_g = GM_{BH}/c^2$ is the gravitational radius and a the spin parameter.

2. Key Jet Parameters

Parameter	Value	Description
M_{BH}	$10^8 M_\odot$	Black hole mass
a_{spin}	0.9	Dimensionless spin
$B_{horizon}$	10^4 T	Magnetic field at BH
Γ_{jet}	10	Lorentz factor

3. Lorentz Factor and Poynting Flux

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}, \quad S = \frac{cB^2}{\mu_0}$$

4. Hoop Stress (Jet Collimation)

$$\sigma_{hoop} = \frac{B_{toroidal}^2}{\mu_0}$$

The toroidal field provides the pinch force collimating the jet.

5. UQFF Jet Modulation

$$g_{jet,UQFF} = P_{BZ}\,(1 - \rho_{SCm}/\rho_{UA})(1 - f_{TRZ})$$

The vacuum aether acts as a slight resistive medium, reducing net jet power by $\sim \rho_{SCm}/\rho_{UA} = 0.1$ in UQFF framework.

6. Observational Links

M87 jet ($P_{BZ} \sim 10^{37}$ W), NGC 1316 (Fornax A radio lobes), Cygnus A (most powerful known jet).

References

- Blandford & Znajek (1977), MNRAS, 179, 433
- UQFF Framework, Star-Magic Session 174

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